

Discrete Time Signal Processing

Discrete Fourier Transform (DFT) & Fast Fourier Transform (FFT)

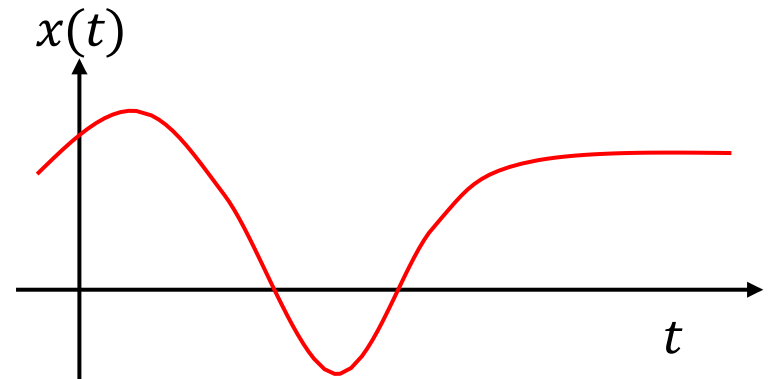
Definition of DFT and IDFT



Signal

Continuous-Time Signals

- Real world signals
- Exists at all times
- Most general form
- E.g. voltage, velocity, vibration



Discrete-Time Signals

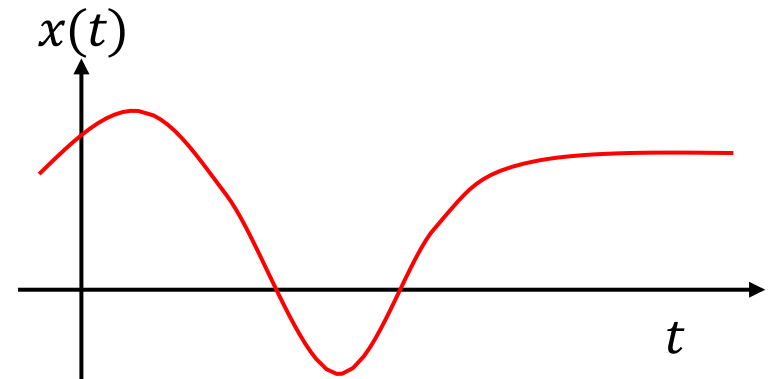
- Processed signals
 - Exists at certain times
 - Preferred form
 - E.g. pixels, daily stock price
-
- **Sampled continuous signal**

where, T_s is sampling time
 k is sample index

Signal

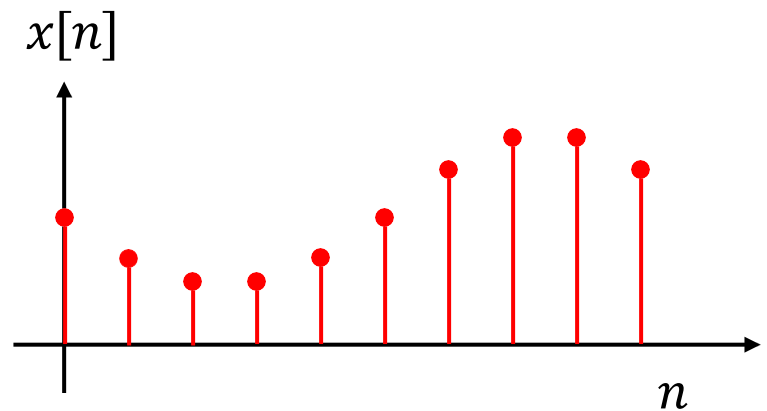
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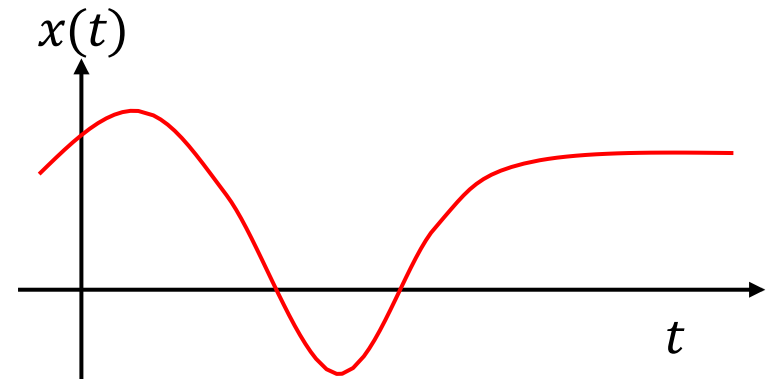


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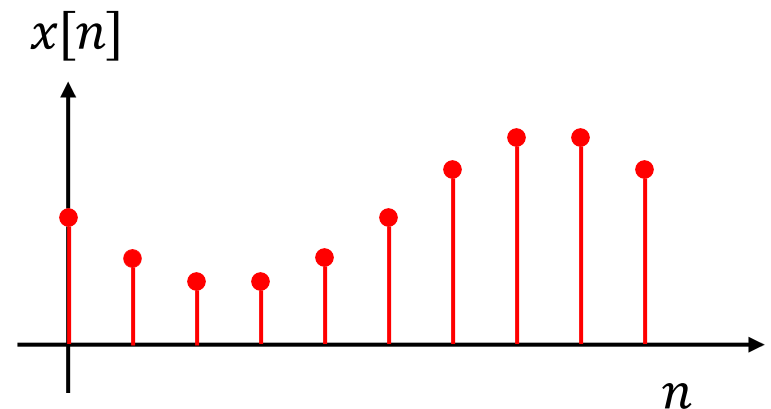
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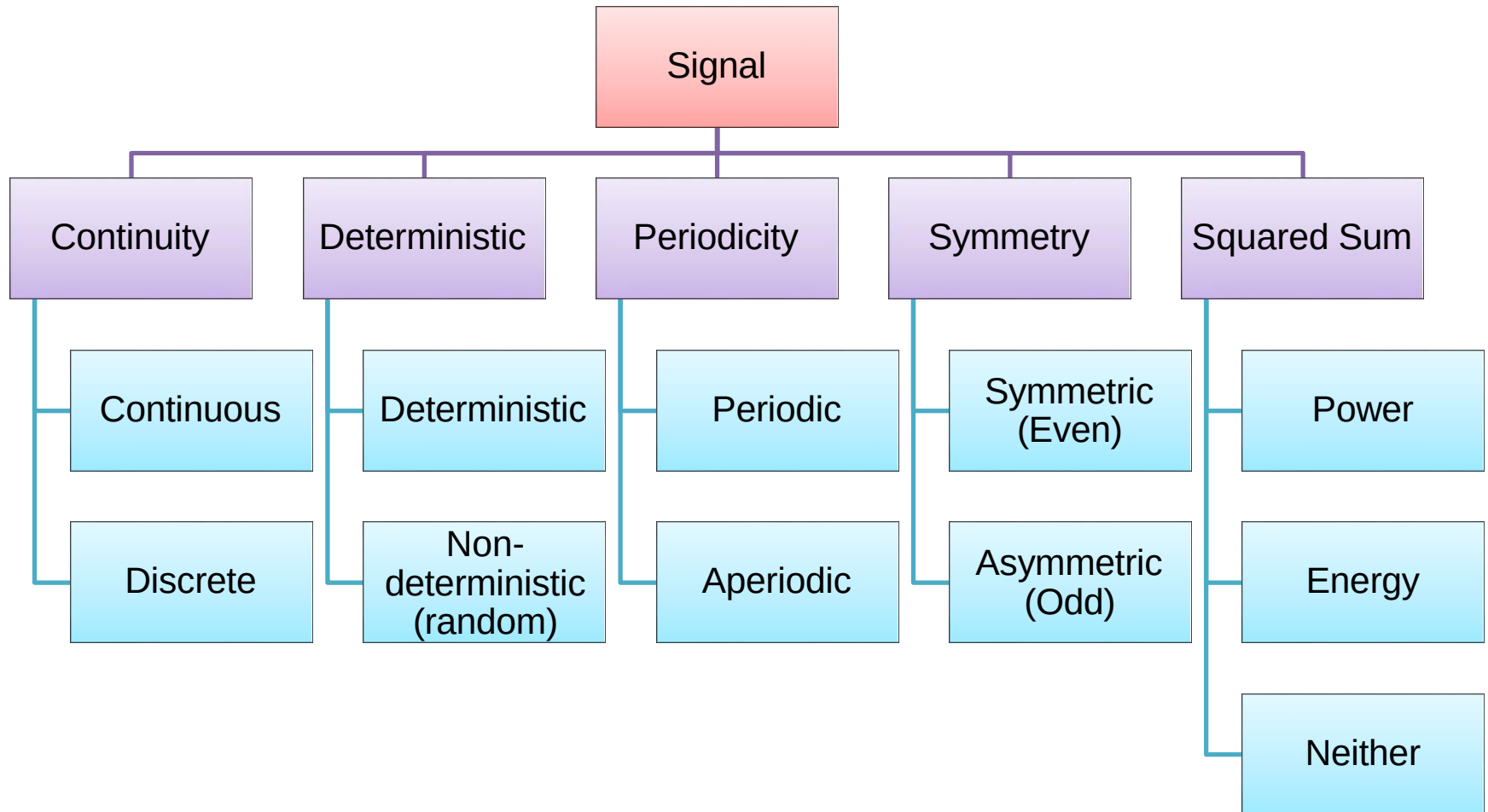
$$x[n] = x(k * T_s)$$

where, T_s is sampling time

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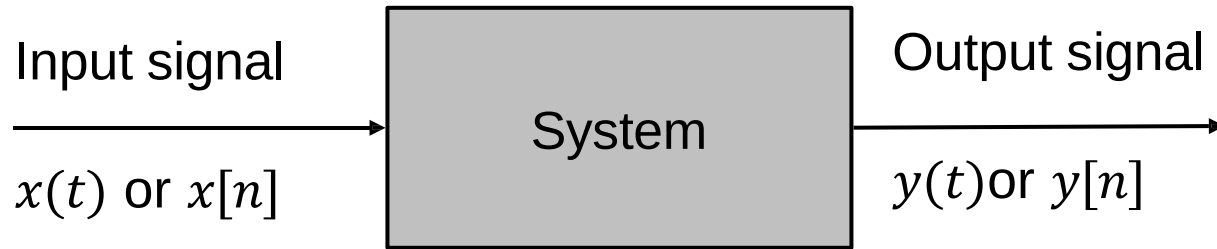


Characteristics of Signal



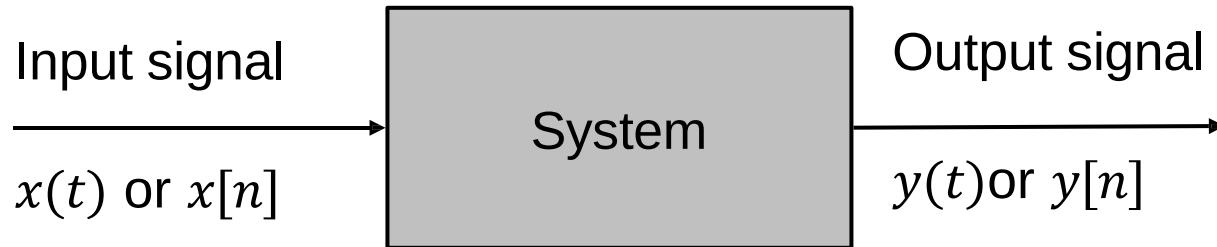
System

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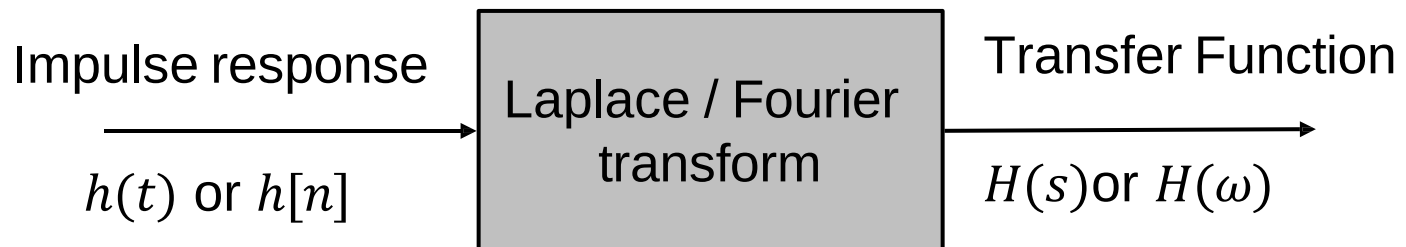


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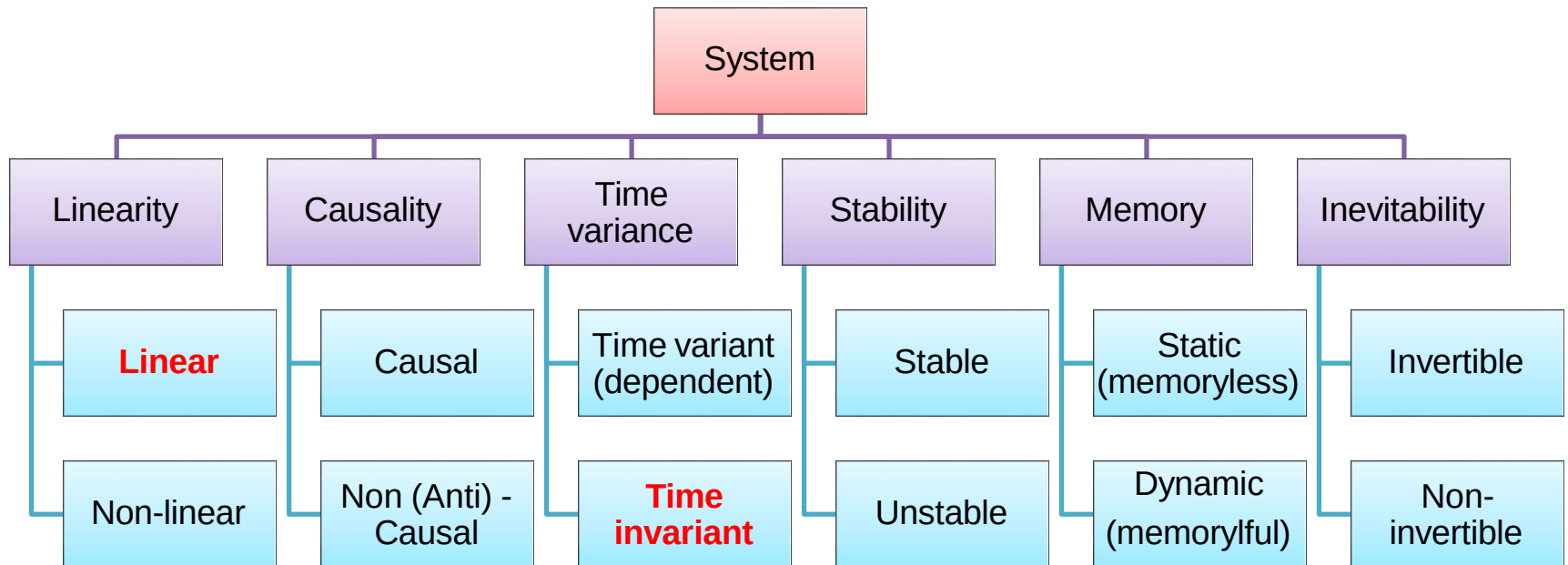
- A system takes a signal as an input and **transforms** it into another signal.



- Characterized by **Impulse Response**: response to impulse signal



Characteristics of System



Remember: **Linear Time-Invariant (LTI)** system

Analog vs Digital Frequency

Analog vs Digital Frequency



Analog frequency



Digital frequency

Analog vs Digital Frequency

Ω

Analog frequency

Range: $(-\infty, \infty)$

ω

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Range: $(-\pi, \pi), (0, 2\pi)$

Analog vs Digital Frequency

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Periodic with 2π

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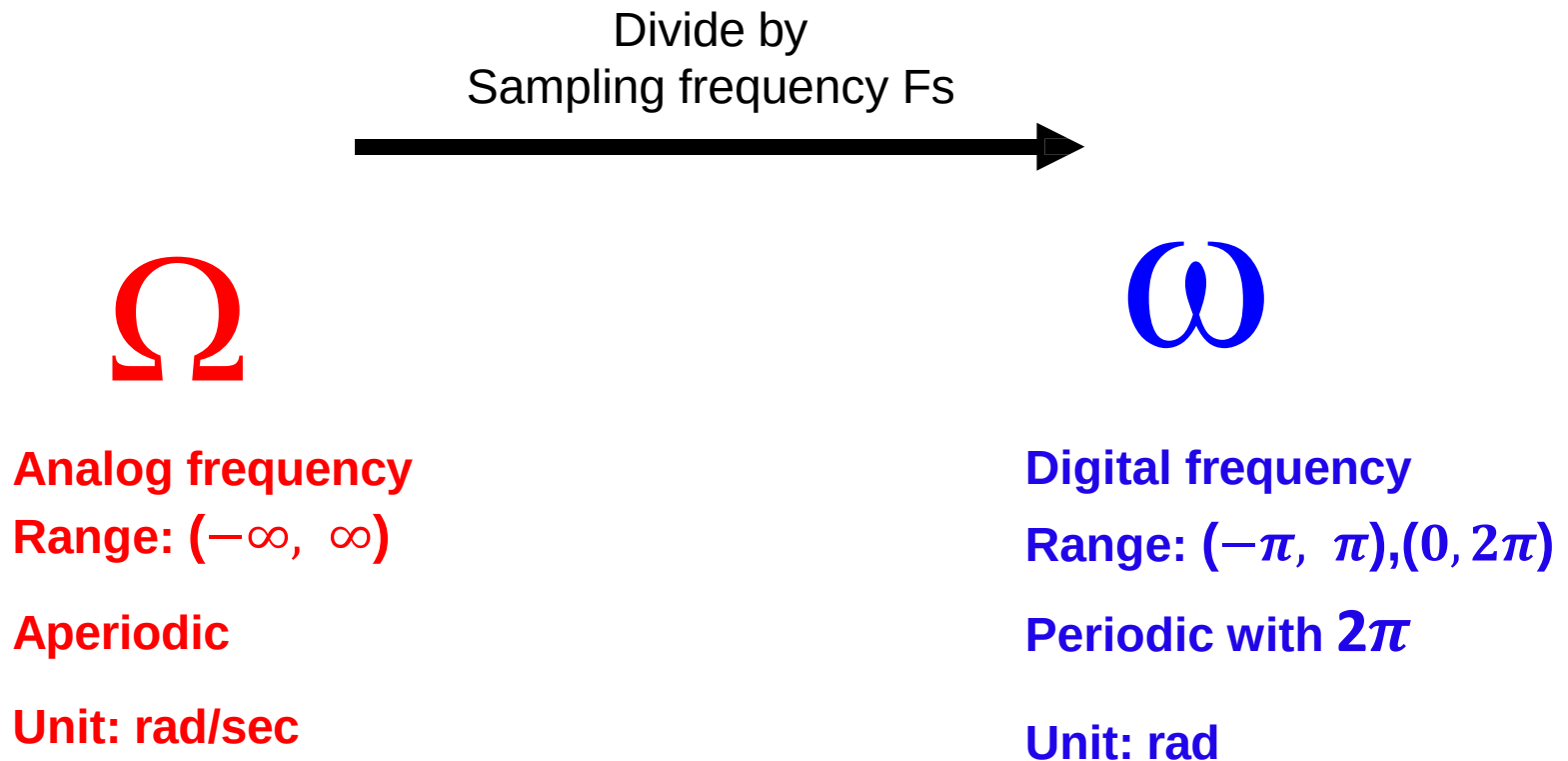
Digital frequency

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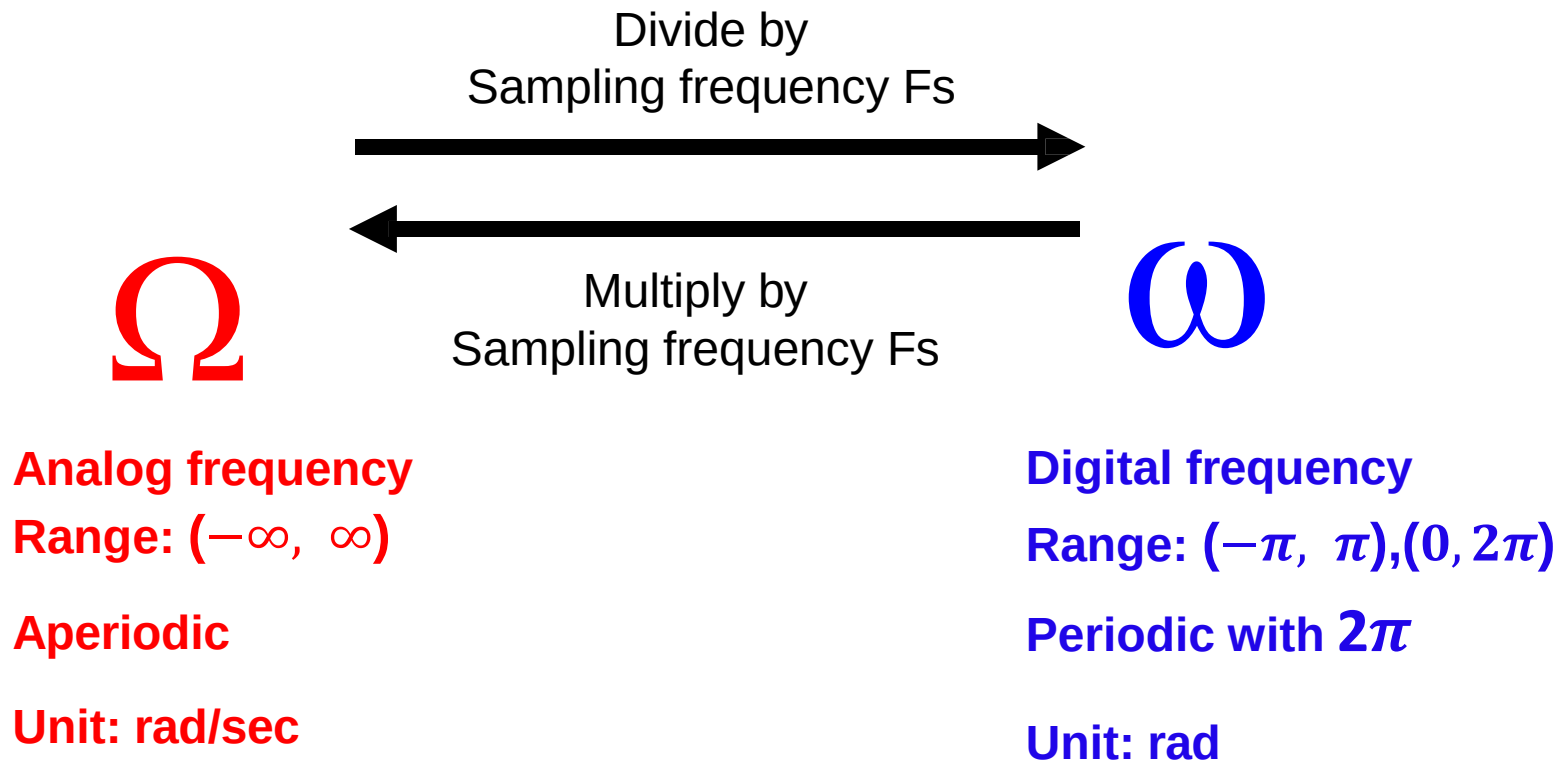
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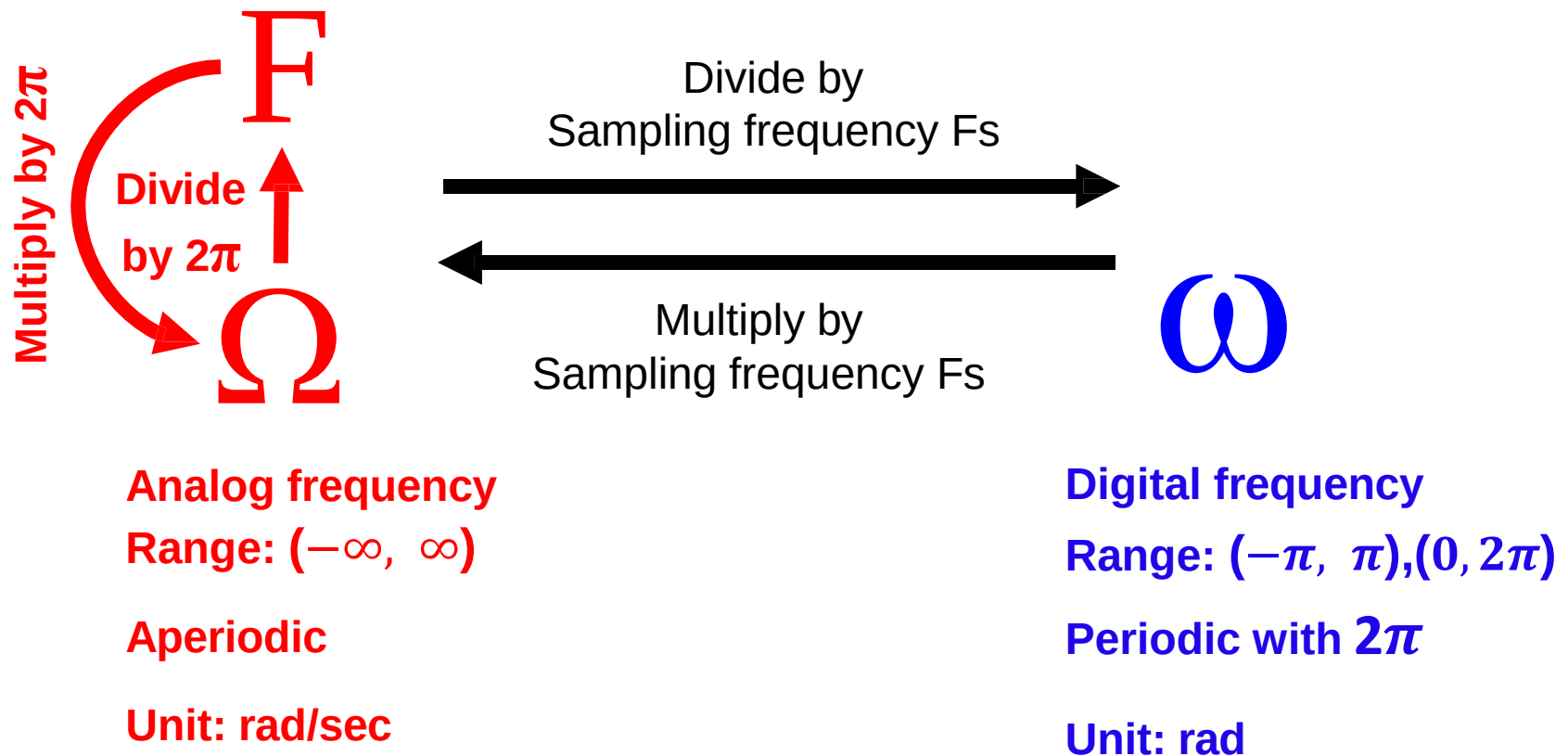
Analog vs Digital Frequency



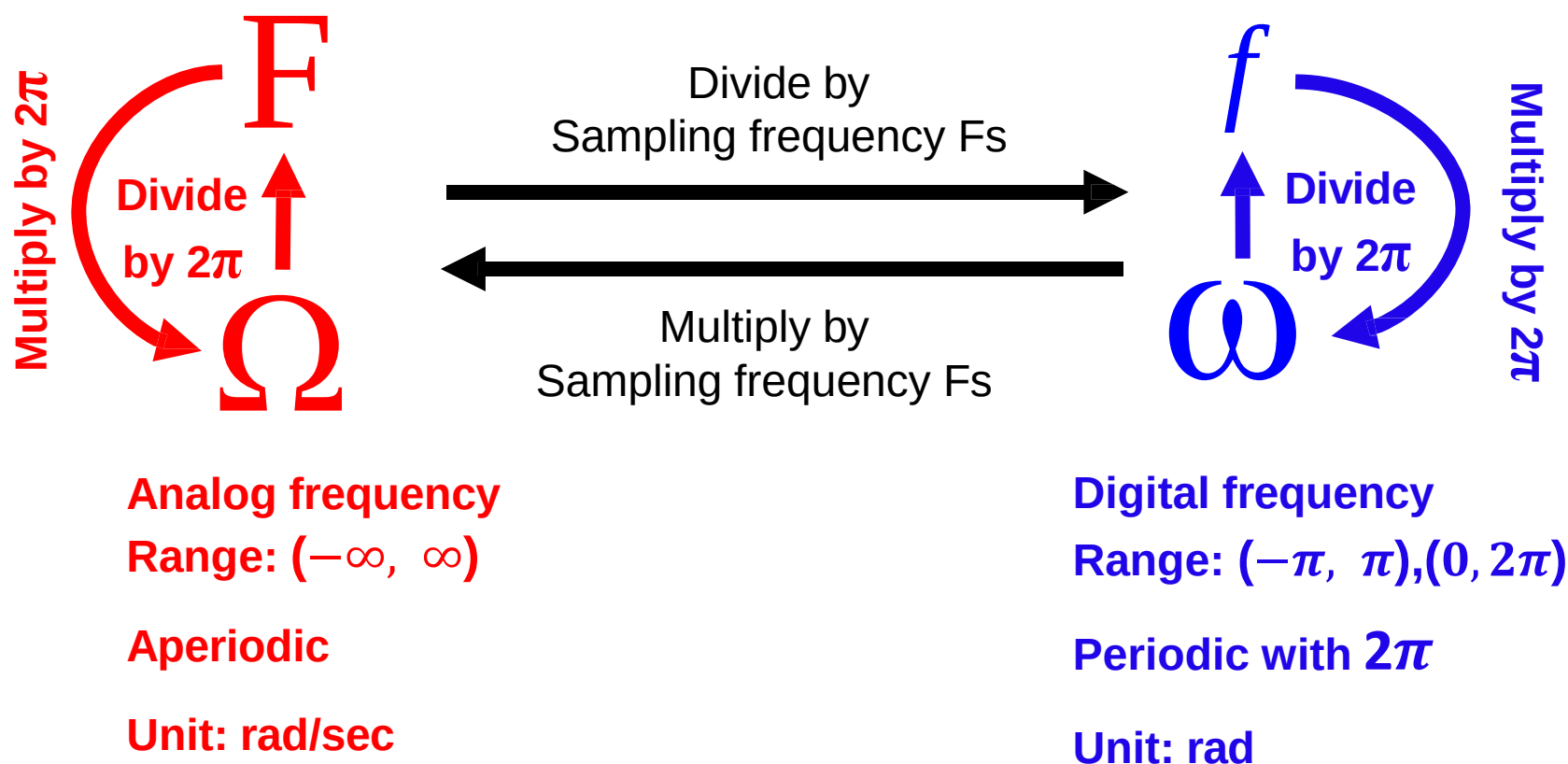
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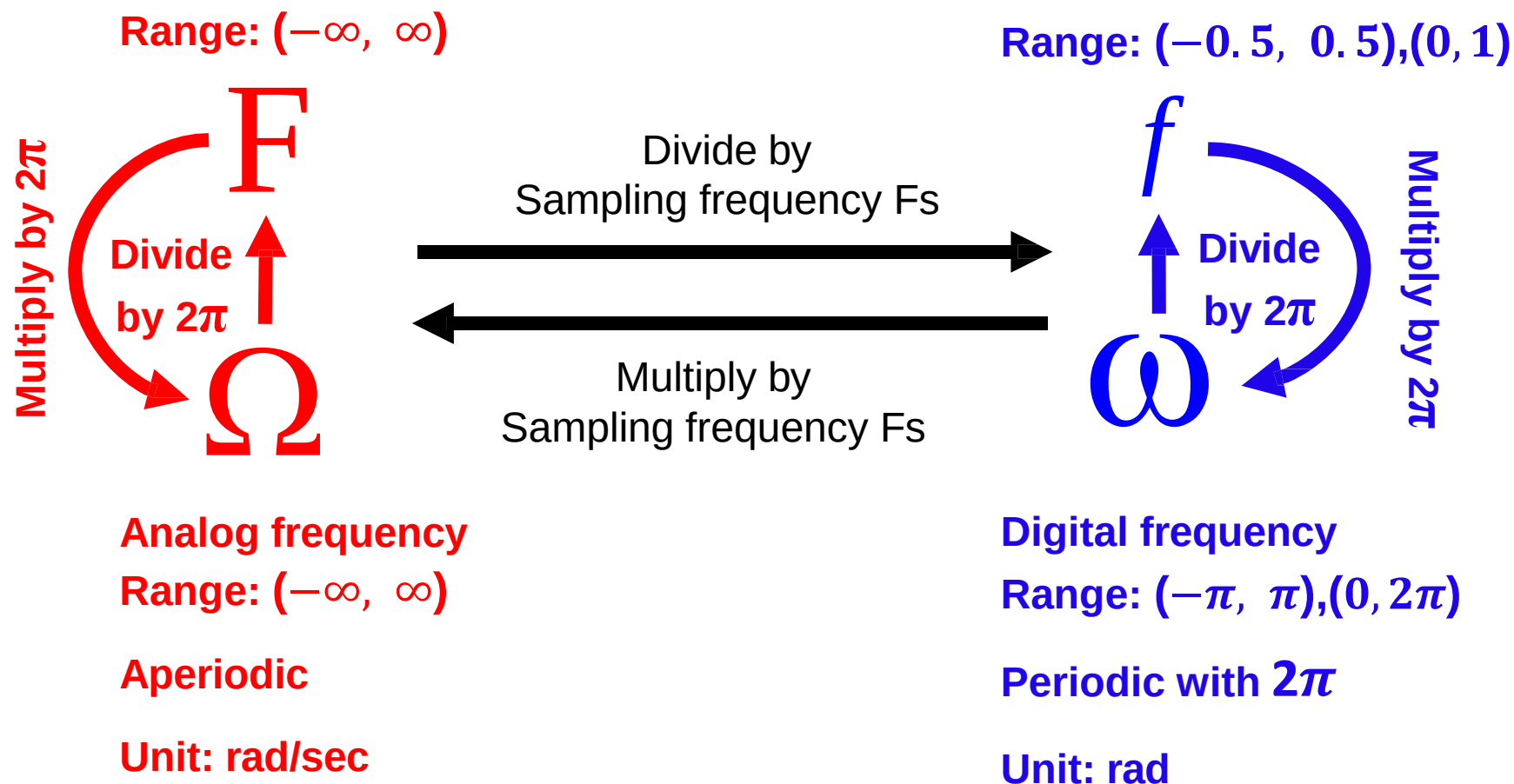
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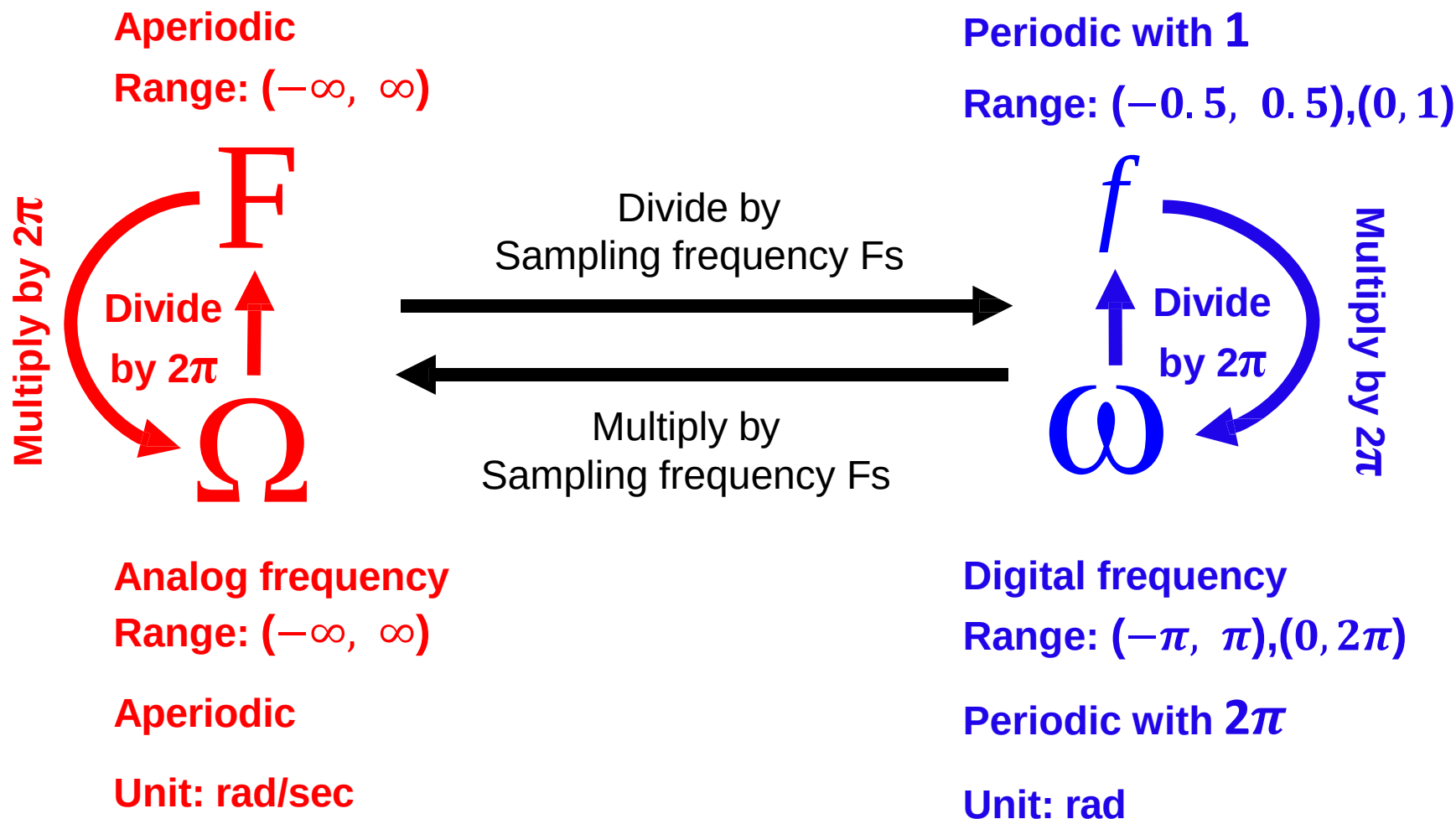
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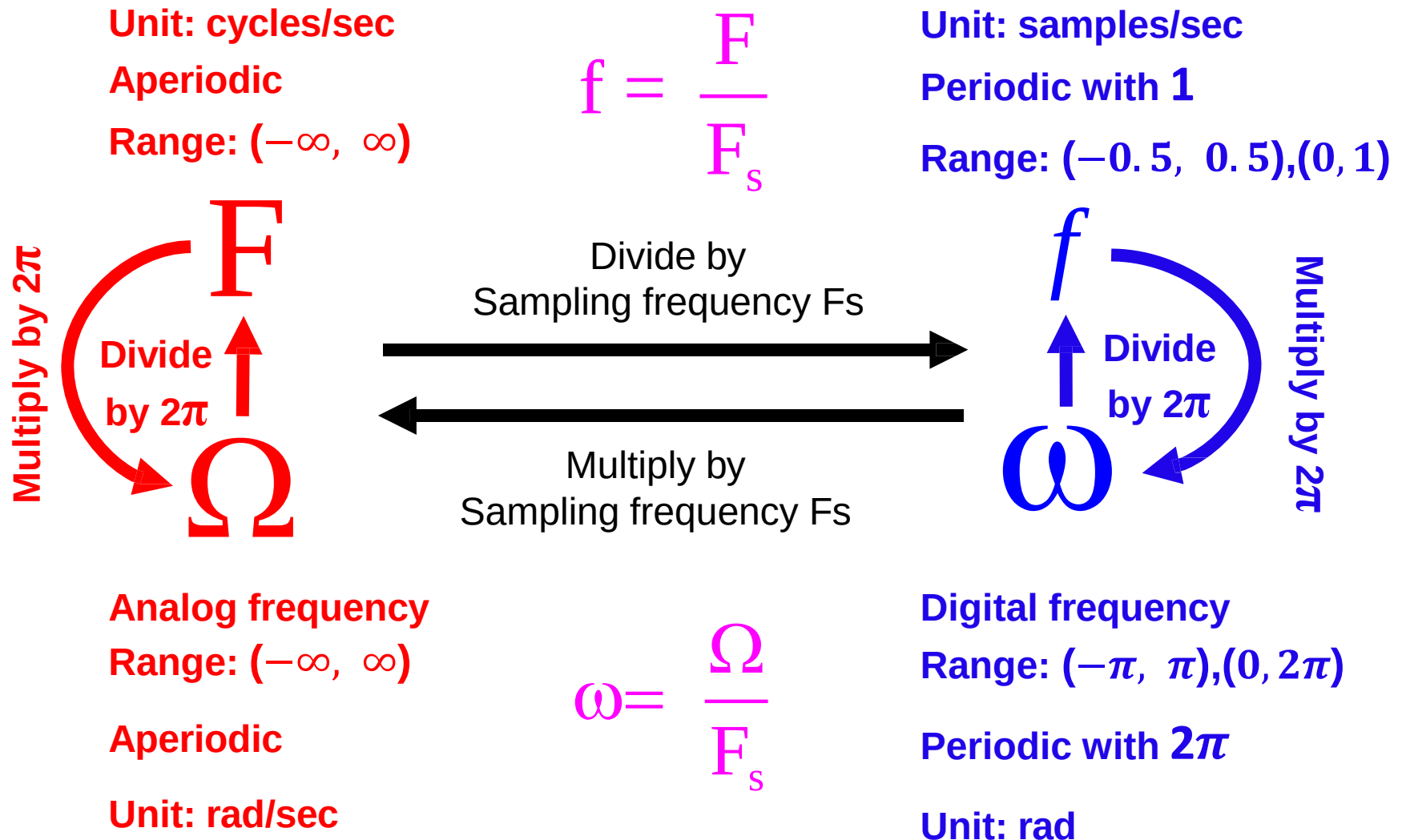
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Review: Discrete Fourier Series

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- Discrete Fourier Series (DFS) represents a **periodic discrete time signal** in terms of **fundamental frequency** and its harmonics:

$$x[n] = \sum_{k=0}^{N-1} C_k e^{jk\omega_0 n} \quad \dots(\text{Synthesis})$$

where, ω_0 is fundamental frequency and $k\omega_0$ is kth harmonic

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- Discrete Time Fourier Transform (DTFT) represents a **aperiodic discrete time signal** as a **continuous function of digital frequency**.

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Summary: Fourier transforms

Transform	Time domain	Frequency domain	Kernel function (Analysis)
CFS	Continuous, Periodic	Aperiodic, Discrete	$e^{-j\Omega_0 k t}$
CTFT	Continuous, Aperiodic	Aperiodic, Continuous	$e^{-j\Omega t}$
DFS	Discrete, Periodic	Periodic, Discrete	$e^{-j\omega_0 k n}$
DTFT	Discrete, Aperiodic	Periodic, Continuous	$e^{-j\omega n}$
DFT	Discrete, Periodic extension	Periodic, Discrete	$e^{-j\frac{2\pi}{N} k n}$

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DFT is un-normalized DFS i.e. scaled by N

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DFT is samples version of DTFT by $2\pi T_N$

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$$\sum_{m=0}^{N-1} \sum_{k=0}^{N-1} e^{j \frac{2\pi}{N} k(n-m)} = \sum_{m=0}^{N-1} \left[\frac{1 - e^{j \frac{2\pi}{N} N(n-m)}}{1 - e^{j \frac{2\pi}{N} (n-m)}} \right]$$

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Properties of DFT



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$$W = e^{-j\frac{2\pi}{N}kn}$$

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$$= \begin{bmatrix} e^{-j\frac{2\pi}{N}0.0} & e^{-j\frac{2\pi}{N}0.1} & \dots & e^{-j\frac{2\pi}{N}k.(N-1)} \\ e^{-j\frac{2\pi}{N}1.0} & \ddots & \vdots & -j2\pi 1.(N-1) \\ \vdots & \dots & e^{-j\frac{2\pi}{N}k.n} & \vdots \\ e^{-j\frac{2\pi}{N}(N-1).0} & e^{-j\frac{2\pi}{N}(N-1).1} & \dots & e^{-j\frac{2\pi}{N}(N-1).(N-1)} \end{bmatrix}$$

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- More precisely,

$$\mathbf{W}_N^{kn} = e^{-j\frac{2\pi}{N}kn}$$

DFT and IDFT

DFT and IDFT

- Discrete Fourier Transform (DFT) represents a **periodic discrete time signal** as a **discrete function of digital frequency**.

$$X[k] = \sum_{n=0}^{N-1} x[n] w_N^{kn} \quad \dots(\text{DFT})$$

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$$W_N^{-kn} = (W_N^{kn})^H$$

Examples-1

Examples-1

Compute DFT of $x[n] = \{1, 2, 3, 4\}$

- Length of input sequence: $N = 4$
- **Using Equation Method:**
- Since, $n, k \in [0, N - 1]$

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$$X[k] = \sum_{n=0}^3 x[n] e^{-j\frac{2\pi}{4}kn} \quad k = 0, 1, 2, 3$$

Examples-1

- Let $k = 0$

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$$X[0] = \sum_{n=0}^3 x[n] e^{-j\frac{2\pi}{4}0n}$$

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$$\begin{aligned} X[0] &= \sum_{n=0}^3 x[n] e^{-j\frac{2\pi}{4}0n} \\ &= x[0] e^{-j\frac{2\pi}{4}00} + x[1] e^{-j\frac{2\pi}{4}01} + x[2] e^{-j\frac{2\pi}{4}02} + x[3] e^{-j\frac{2\pi}{4}03} \end{aligned}$$

Examples-1

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First DFT coefficient is equal to sum of input sequence

Examples-1

- Let $k = 1$

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Examples-1

- Let $k = 1$

$$\begin{aligned} X[1] &= \sum_{n=0}^3 x[n] e^{-j\frac{2\pi}{N}1n} \\ &= x[0] e^{-j\frac{2\pi}{4}10} + x[1] e^{-j\frac{2\pi}{4}11} + x[2] e^{-j\frac{2\pi}{4}12} + x[3] e^{-j\frac{2\pi}{4}13} \\ &= 1.1 + 2 e^{-j\frac{\pi}{2}} + 3 e^{-j\pi} + 4 e^{-j\frac{3\pi}{4}} \end{aligned}$$

Examples-1

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Examples-1

- Let $k = 2$

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$$\begin{aligned} X[2] &= \sum_{n=0}^3 x[n] e^{-j\frac{2\pi}{N}2n} \\ &= x[0] e^{-j\frac{2\pi}{4}2\cdot 0} + x[1] e^{-j\frac{2\pi}{4}2\cdot 1} + x[2] e^{-j\frac{2\pi}{4}2\cdot 2} + x[3] e^{-j\frac{2\pi}{4}2\cdot 3} \end{aligned}$$

Examples-1

- Let $k = 2$

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Examples-1

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Examples-1

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Examples-1

Examples-1

- For input sequence

$$x[n] = \{1, 2, 3, 4\}$$

- 4-point DFT is

Examples-1

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$$X[k] = \{10, -2 + j2, -2, -2 - j2\}$$

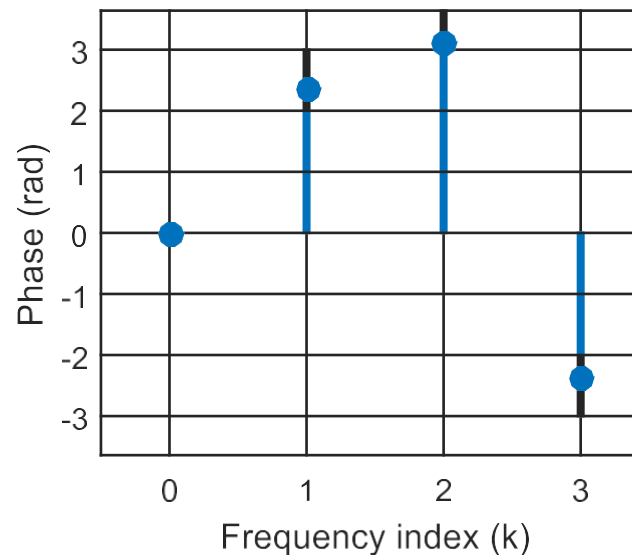
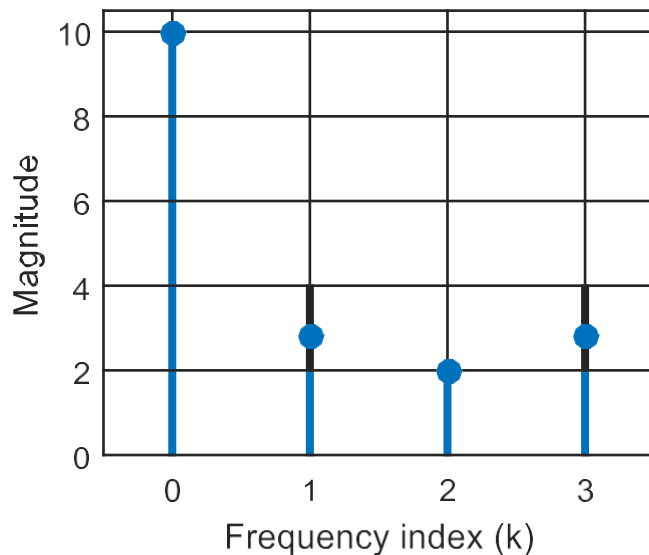
Examples-1

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Examples-1

- **Using Matrix Method:**

Examples-1

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$$\mathbf{X} = \mathbf{W}\mathbf{x}$$

Examples-1

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$$\mathbf{W}_4 = \begin{bmatrix} e^{-j\frac{2\pi}{4}0.0} & e^{-j\frac{2\pi}{4}0.1} & e^{-j\frac{2\pi}{4}0.2} & e^{-j\frac{2\pi}{4}0.3} \\ e^{-j\frac{2\pi}{4}1.0} & e^{-j\frac{2\pi}{4}1.1} & e^{-j\frac{2\pi}{4}1.2} & e^{-j\frac{2\pi}{4}1.3} \\ e^{-j\frac{2\pi}{4}2.0} & e^{-j\frac{2\pi}{4}2.1} & e^{-j\frac{2\pi}{4}2.2} & e^{-j\frac{2\pi}{4}2.3} \\ e^{-j\frac{2\pi}{4}3.0} & e^{-j\frac{2\pi}{4}3.1} & e^{-j\frac{2\pi}{4}3.2} & e^{-j\frac{2\pi}{4}3.3} \end{bmatrix}$$

Examples-1

Examples-1

$$\mathbf{W}_4 = \begin{bmatrix} e^0 & e^0 & e^0 & e^0 \\ e^0 & e^{-j\frac{\pi}{2}} & e^{-j\pi} & e^{-j\frac{3\pi}{2}} \\ e^0 & e^{-j\pi} & e^{-j2\pi} & e^{-j3\pi} \\ e^0 & e^{-j\frac{3\pi}{2}} & e^{-j3\pi} & e^{-j\frac{\pi}{2}} \end{bmatrix}$$

Examples-1

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$$\mathbf{W}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

Examples-1

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4-point DFT matrix

Examples-1

Examples-1

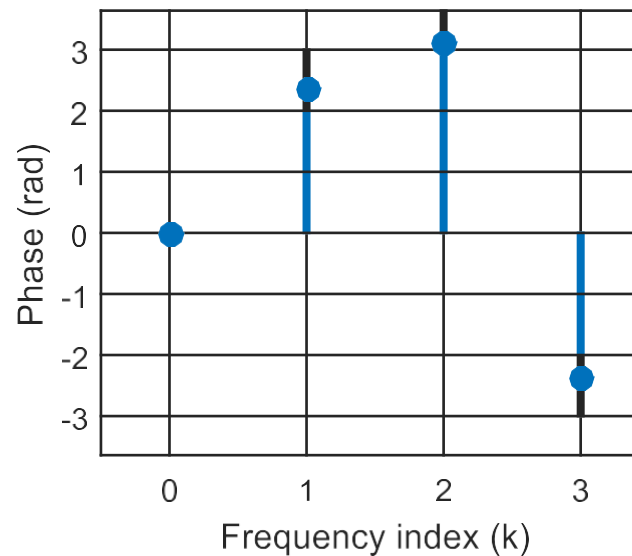
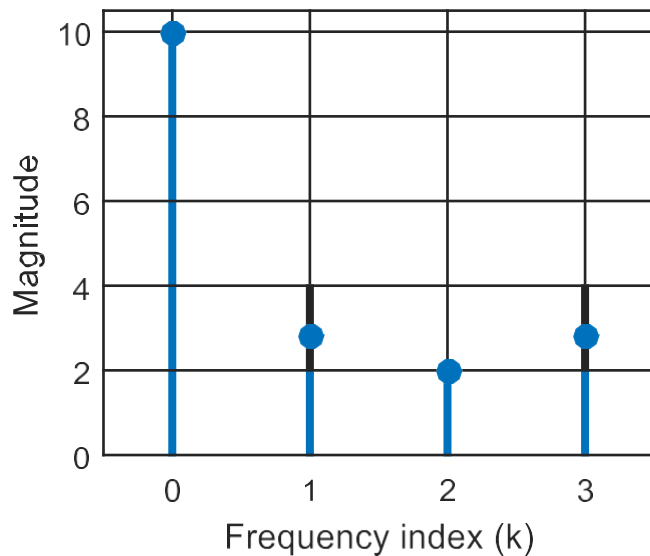
$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

Examples-1

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Properties of DFT

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If $x_1[n] \xrightarrow{DFT} X_1[k]$ and $x_2[n] \xrightarrow{DFT} X_2[k]$

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If $x_1[n] \xrightarrow{DFT} X_1[k]$ and $x_2[n] \xrightarrow{DFT} X_2[k]$

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$$X_1[k] = \sum_{n=0}^{N-1} x_1[n] w_N^{kn} \quad \text{and} \quad X_2[k] = \sum_{n=0}^{N-1} x_2[n] w_N^{kn}$$

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$$X_3[k] = \sum_{n=0}^{N-1} (a x_1[n] + b x_2[n]) w_N^{kn}$$

Properties of DFT

▪ Linearity

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$$\begin{aligned} X_3[k] &= \sum_{n=0}^{N-1} (a x_1[n] + b x_2[n]) w_N^{kn} \\ &= \sum_{n=0}^{N-1} a x_1[n] w_N^{kn} + \sum_{n=0}^{N-1} b x_2[n] w_N^{kn} \end{aligned}$$

Properties of DFT

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Properties of DFT

Properties of DFT

- **Circular shift**

If $x[n]$ $\longrightarrow$ $|$
then

Properties of DFT

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$$\begin{array}{l} \text{If } x[n] \longrightarrow | \\ \text{then } x[(n-m)_N] \longrightarrow e^{j2\pi k m / N} X[k] \end{array}$$

Properties of DFT

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Properties of DFT

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Properties of DFT

Properties of DFT

- **Modulation or Frequency shift**

If $x[n]$ $\longrightarrow$ $|$
then

Properties of DFT

- Modulation or Frequency shift

$$\begin{array}{l} \text{If } x[n] \\ \text{then } e^{-j\left(\frac{2\pi m}{N}\right)n} x[n] \xrightarrow{DFT} X[(k+m)_N] \end{array}$$

Properties of DFT

- **Modulation or Frequency shift**

If $x[n]$
then $e^{-j(2\pi m/N)n} x[n] \xrightarrow{DFT} X[(k+m)_N]$

$$X[k] = \sum_{n=0}^{N-1} e^{-j(2\pi m/N)n} x[n] w_N^{kn}$$

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$$\begin{aligned} X[k] &= \sum_{n=0}^{N-1} e^{-j(2\pi m/N)n} x[n] w_N^{kn} \\ &= \sum_{n=0}^{N-1} w_N^{mn} x[n] w_N^{kn} \quad \dots \left(w_N = e^{-j2\pi/N} \right) \end{aligned}$$

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- Modulation or Frequency shift

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Properties of DFT

[Source: Discrete Time Signal Processing, Alan Oppenheim]

Properties of DFT

$$x_1[n]x_2[n]$$

$$\frac{1}{N} \sum_{\ell=0}^{N-1} X_1(\ell) X_2[((k-\ell))_N]$$

$$x^*[n]$$

$$X^*[((-k))_N]$$

$$x^*[((-n))_N]$$

$$X^*[k]$$

$$\mathcal{R}e\{x[n]\}$$

$$X_{\text{ep}}[k] = \frac{1}{2}\{X[((k))_N] + X^*[(((-k))_N)]\}$$

$$j\mathcal{I}m\{x[n]\}$$

$$X_{\text{op}}[k] = \frac{1}{2}\{X[((k))_N] - X^*[(((-k))_N)]\}$$

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$$j\mathcal{Im}\{x[n]\}$$

$$X_{\text{op}}[k] = \frac{1}{2} \{X[((k))_N] - X^*[(((-k))_N)]\}$$

$$x_{\text{ep}}[n] = \frac{1}{2} \{x[n] + x^*[(((-n))_N)]\}$$

$$\mathcal{Re}\{X[k]\}$$

$$x_{\text{op}}[n] = \frac{1}{2} \{x[n] - x^*[(((-n))_N)]\}$$

$$j\mathcal{Im}\{X[k]\}$$

- If $x[n]$ is real then

Symmetry properties

$$\left\{ \begin{array}{l} X[k] = X^*[(((-k))_N)] \\ \mathcal{Re}\{X[k]\} = \mathcal{Re}\{X[(((-k))_N)]\} \\ \mathcal{Im}\{X[k]\} = -\mathcal{Im}\{X[(((-k))_N)]\} \\ |X[k]| = |X[(((-k))_N)]| \\ \angle\{X[k]\} = -\angle\{X[(((-k))_N)]\} \end{array} \right.$$

[Source: Discrete Time Signal Processing, Alan Oppenheim]

Example-2



Example-2

Circular shift by -1 or +3

Circular shift by +1 or -3

Sum of $x_3[n]$ and $x_4[n]$



Convolution of $x_1[n]$ and $x_2[n]$

Example-2

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Example-2

Compute DFT of $x_1[n] = \{1, 2, 3, 4\}$ and $x_2[n] = \{1, 0, 2, -2\}$ using the results not otherwise find DFT of following (MAY19, 10 marks)

$$x_3[n] = \{4, 1, 2, 3\}$$

Circular shift by -1 or +3

$$x_4[n] = \{2, 3, 4, 1\}$$

Circular shift by +1 or -3

$$x_5[n] = \{6, 4, 6, 4\}$$

Sum of $x_3[n]$ and $x_4[n]$

$$x_6[n] = x_1[n] \otimes x_2[n]$$

Convolution of $x_1[n]$ and $x_2[n]$

- Length of input sequence: $N = 4$
- **Using Matrix method:**

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$$\mathbf{X}_2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \\ -2 \end{bmatrix}$$

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$$x_3[n] = \{4, 1, 2, 3\}$$

- Using circular shift property

$$x[(n-1)_4] \longrightarrow e^{j2\pi k/4} X[k]$$

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Example-2

Example-2

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Example-2

$$x_5[n] = \{6, 4, 6, 4\}$$

- Using linearity property

$$x_3[n] + x_4[n] \xrightarrow{DFT} X_3[k] + X_4[k]$$

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$$x_3[n] + x_4[n] \xrightarrow{DFT} X_3[k] + X_4[k]$$

$$\mathbf{X}_5 = \begin{bmatrix} 10 \\ 2 + j2 \\ 2 \\ 2 - j2 \end{bmatrix} + \begin{bmatrix} 10 \\ -2 - j2 \\ 2 \\ -2 + j2 \end{bmatrix}$$

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$$x_3[n] + x_4[n] \xrightarrow{DFT} X_3[k] + X_4[k]$$

$$\mathbf{X}_5 = \begin{bmatrix} 10 \\ 2 + j2 \\ 2 \\ 2 - j2 \end{bmatrix} + \begin{bmatrix} 10 \\ -2 - j2 \\ 2 \\ -2 + j2 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \\ 4 \\ 0 \end{bmatrix}$$

Circular convolution using DFT and IDFT



Properties of DFT

Properties of DFT

- **Convolution**

If $x_1[n] \xrightarrow{DFT} X_1[k]$ and $x_2[n] \xrightarrow{DFT} X_2[k]$

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Properties of DFT

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If $x_1[n] \xrightarrow{DFT} X_1[k]$ and $x_2[n] \xrightarrow{DFT} X_2[k]$

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Properties of DFT

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Circular Convolution using DFT

Circular Convolution using DFT

- Let $x_1[n]$ and $x_2[n]$ be two signals of length N_1 and N_2
- Calculate length of circular convolution output: $N = \max(N_1, N_2)$
- Calculate N-point DFT of both sequences.
- Multiply both DFT sequences sample-by-sample to calculate DFT of circular convolution output.

where

- Compute IDFT of $X_3[k]$ to get time-domain representation of circularly convolved sequence.

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$$X_3[k] = X_1[k] \cdot X_2[k] \quad k = 0, 1, \dots, N-1$$

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Example-2

$$x_6[n] = x_1[n] \otimes x_2[n]$$

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$$\mathbf{X}_6 = \begin{bmatrix} 10 \\ -2 + j2 \\ -2 \\ -2 - j2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 - j2 \\ 5 \\ -1 + j2 \end{bmatrix}$$

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$$\mathbf{x}_6 = \begin{bmatrix} 1 & -2 & 2 & 0 \\ 0 & 1 & -2 & 2 \\ 2 & 0 & 1 & -2 \\ -2 & 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

**Circular convolution
using matrix method**

Example-2

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DFT using
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DFT using
matrix method

Example-3

Example-3

Compute circular convolution of following sequences using DFT and IDFT
 $x_1[n] = \{1, 2, 1, 2\}$ and $x_2[n] = \{1, 2, 1\}$ (DEC18)

- Length of convolution output sequence: $N = \max(4, 3) = 4$
- **Zero pad $x_2[n]$ to make length of signal equal to N.**
$$x_2[n] = \{1, 2, 1, 0\}$$

- **Using Matrix method:**

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Calculate 4-point DFT of both sequences.

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Example-3

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Calculate 4-point DFT of both sequences.

$$\mathbf{X}_1 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix}$$

$$\mathbf{X}_2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$$

Example-3

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Example-3



Example-3

$$x_3[n] = x_1[n] \otimes x_2[n]$$

- Using convolution property

$$x_1[n] \otimes x_2[n] \xrightarrow{DFT} X_1[k] \cdot X_2[k]$$

Example-3

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Example-3

- Compute IDFT of $X_3(k)$ to get time-domain representation of circularly convolved sequence

Example-3

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$$x[n] = \frac{1}{N} \sum_{n=0}^{N-1} X[k] W_N^{-kn} \quad \dots(\text{IDFT})$$

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4-point DFT

Example-3

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$$W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \quad (W_4)^H = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

4-point DFT

4-point IDFT

Example-3

Example-3

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

Example-3

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Example-3

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**Circular convolution
using matrix method**

Example-3

$$\mathbf{X}_3 = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 24 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \\ 6 \\ 6 \end{bmatrix}$$

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**Circular convolution
using matrix method**

Example-4



Example-4

Circular shift by -2 or +2

Sum of $x_3[n]$ and $x_2[n]$

$\otimes$

Convolution of $x_1[n]$ and $x_2[n]$

Example-4

Circular shift by -2 or +2

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Circular shift by -2 or +2

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Example-4

Compute DFT of $x_1[n] = \{-1, 2, -1, -2\}$ and $x_2[n] = \{1, 2, -1\}$ using the results not otherwise find DFT of following

$$x_3[n] = \{-1, -2, -1, 2\} \quad \text{Circular shift by -2 or +2}$$

$$x_4[n] = \{0, 0, -2, 2\} \quad \text{Sum of } x_3[n] \text{ and } x_2[n]$$

$$x_5[n] = x_1[n] \otimes x_2[n] \quad \text{Convolution of } x_1[n] \text{ and } x_2[n]$$

- Length of input sequence: $N = 4$
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Example-4

Example-4

$$\mathbf{X}_1 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ -1 \\ -2 \end{bmatrix}$$

Example-4

$$\mathbf{X}_1 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ -1 \\ -2 \end{bmatrix} = \begin{bmatrix} -2 \\ -j4 \\ -2 \\ j4 \end{bmatrix}$$

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Example-4

Example-4

$$x_3[n] = \{-1, -2, -1, 2\}$$

- Using circular shift property

$$x[(n-2)_4] \longrightarrow e^{j2\pi k/4} X[k]$$

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Example-4

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Example-4

$$x_4[n] = \{0, 0, -2, 2\}$$

- Using linearity property

$$x_2[n] + x_3[n] \xrightarrow{DFT} X_2[k] + X_3[k]$$

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Example-4

$$x_5[n] = x_1[n] \otimes x_2[n]$$

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Overlap-save and Overlap-add



Review: Example-3

Compute circular convolution of following sequences using DFT and IDFT
 $x_1[n] = \{1, 2, 1, 2\}$ and $x_2[n] = \{1, 2, 1\}$

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What if linear convolution?

Linear convolution through DFT

Linear convolution through DFT

- Let $x_1[n]$ and $x_2[n]$ be two signals of length N_1 and N_2
- Calculate length of linear convolution output: $N = N_1 + N_2 - 1$
- Zero pad signals to achieve required length.
- Calculate N-point DFT of both sequences.
- Multiply both DFT sequences sample-by-sample to calculate DFT of circular convolution output.

where

- Compute IDFT of $X_3[k]$ to get time-domain representation of circularly convolved sequence.

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$$X_3[k] = X_1[k] \cdot X_2[k] \quad k = 0, 1, \dots, N-1$$

$$\text{where } x_1[n] \otimes x_2[n] \xrightarrow{\text{DFT}} X_3[k]$$

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- Compute IDFT of $X_3[k]$ to get time-domain representation of circularly convolved sequence.

Example-5

Compute **linear** convolution of following sequences using DFT and IDFT
 $x_1[n] = \{1, 2, 2\}$ and $x_2[n] = \{1, -1\}$

- Length of convolution output sequence: $N = 3 + 2 - 1 = 4$
- **Zero pad $x_1[n]$ and $x_2[n]$ to make length of signal equal to N.**

$$x_1[n] = \{1, 2, 2, 0\}$$

$$x_2[n] = \{1, -1, 0, 0\}$$

- **Using Matrix method:**

Example-5

Calculate 4-point DFT of both sequences.

Example-5

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$$\mathbf{X}_1 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

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Example-5



Example-5

$$x_3[n] = x_1[n] \otimes x_2[n]$$

- Using convolution property

$$x_1[n] \otimes x_2[n] \xrightarrow{DFT} X_1[k] \cdot X_2[k]$$

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$$x_1[n] \otimes x_2[n] \xrightarrow{DFT} X_1[k] \cdot X_2[k]$$

$$\mathbf{X}_3 = \begin{bmatrix} 5 \\ -1 - j2 \\ 1 \\ -1 + j2 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 + j1 \\ 2 \\ 1 - j1 \end{bmatrix}$$

Example-5

$$x_3[n] = x_1[n] \otimes x_2[n]$$

- Using convolution property

$$x_1[n] \otimes x_2[n] \xrightarrow{DFT} X_1[k] \cdot X_2[k]$$

$$\mathbf{X}_3 = \begin{bmatrix} 5 \\ -1 - j2 \\ 1 \\ -1 + j2 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 + j1 \\ 2 \\ 1 - j1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 - j3 \\ 2 \\ 1 + j3 \end{bmatrix}$$

Example-5

Example-5

$$\mathbf{X}_3 = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 0 \\ 1-j3 \\ 2 \\ 1+j3 \end{bmatrix}$$

Example-5

$$\mathbf{X}_3 = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 0 \\ 1-j3 \\ 2 \\ 1+j3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -2 \end{bmatrix}$$

Example-5

$$\mathbf{X}_3 = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 0 \\ 1-j3 \\ 2 \\ 1+j3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -2 \end{bmatrix}$$

$$\mathbf{X}_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

Linear convolution
using matrix method

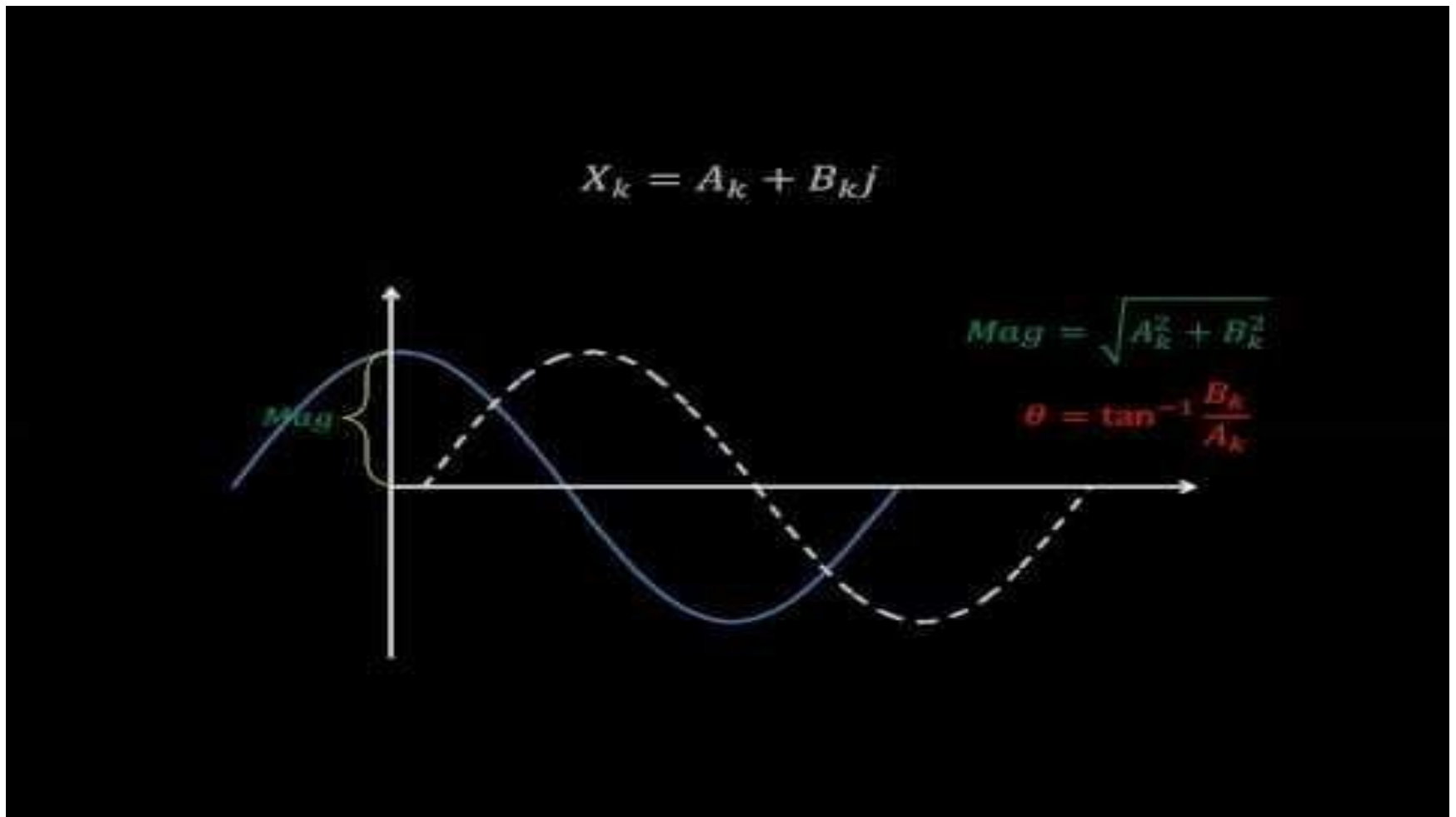
Example-5

$$\mathbf{X}_3 = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 0 \\ 1-j3 \\ 2 \\ 1+j3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -2 \end{bmatrix}$$

$$\mathbf{X}_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -2 \end{bmatrix}$$

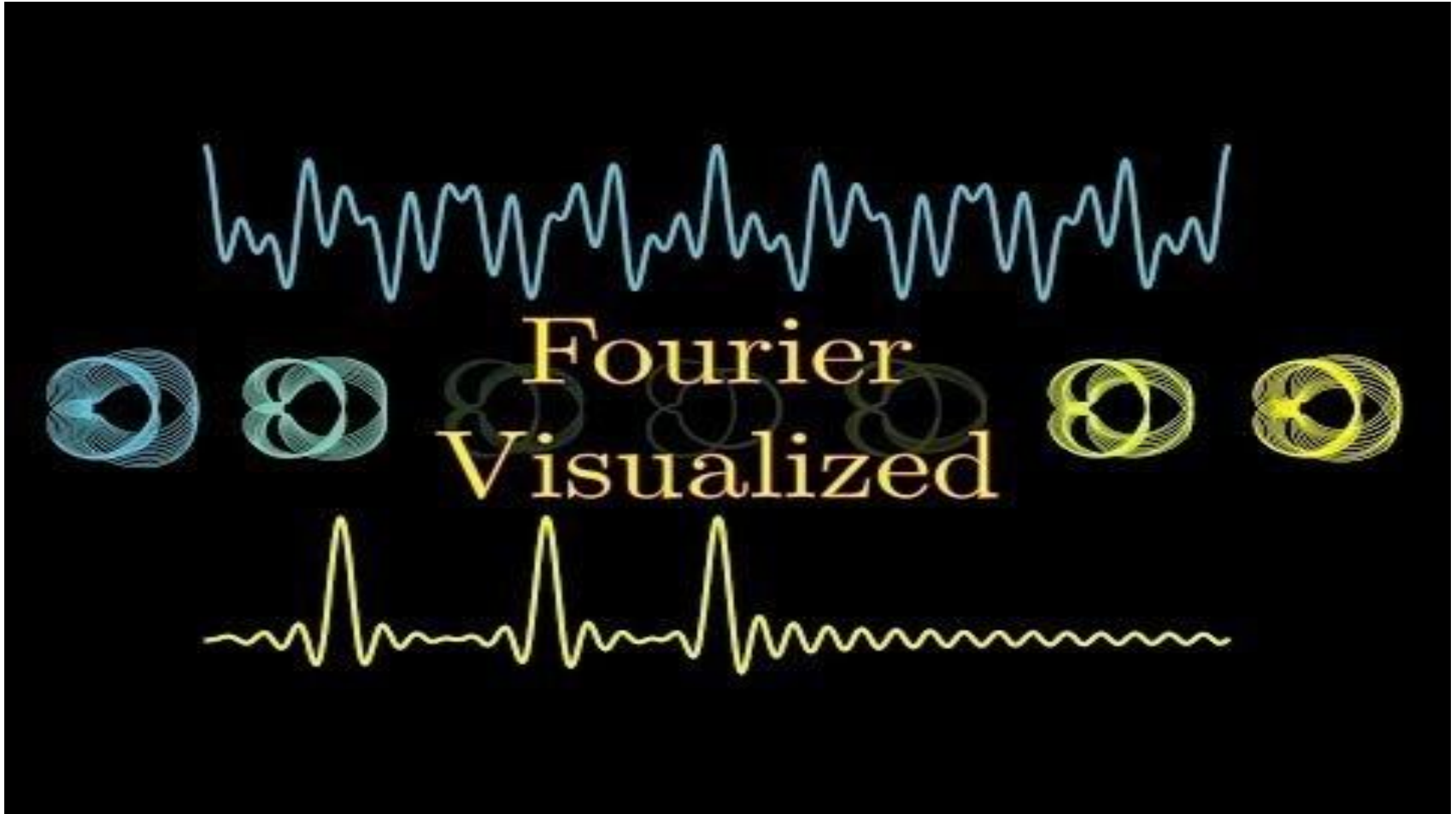
Linear convolution
using matrix method

DFT: Step-by-step



[Discrete Fourier Transform: Step-by-Step](#)

Fourier Visualized



[Fourier Visualization](#)

Thank You